

- Q.19 Explain the construction details and working of capacitive microphone.
- Q.20 Explain the working of shaft encoder. Write any two applications of it.
- Q.21 Explain the construction details and working principle of inductive microphone.
- Q.22 What is seismic transducer? What are the applications of it?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 What is piezoelectric transducer? Explain the working of it. Write down any two applications of piezoelectric transducer.
- Q.24 Explain the working principle of humidity sensor. Write advantages and disadvantages of it. How does an LDR work?
- Q.25 Draw the diagram of LVDT. Explain its working. How it is different from RVDT? Write down its applications.

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3rd Sem / Instrumentation and Control

Subject : Sensors and Transducers

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which of the following is an example of secondary transducers
- a) Thermocouple
 - b) Shaft encoder
 - c) Piezoelectric transducer
 - d) LVDT
- Q.2 Potentiometer is a
- a) Capacitive transducer
 - b) Resistive transducer
 - c) Inductive transducer
 - d) Digital transducer

Q.3 An RVDT works on the principle of

- a) Electromagnetic induction
- b) Mutual induction
- c) Seebeck effect
- d) None of the above

Q.4 If we apply a sunlight on LDR then

- a) Resistance changes
- b) Resistance does not change.
- c) Pressure changes.
- d) All of the above

Q.5 Piezoelectric transducer is an example of

- a) Analog transducer b) Active transducer
- c) Inverse transducer d) All of the above

Q.6 Which of the following is a dynamic characteristics of transducer?

- a) Hysteresis b) Sensitivity
- c) Resolution d) Response time

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Write the principle of operation of potentiometer.

Q.8 Expand LVDT.

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Q.9 RVDT works on the principle of self inductance. (True/False)

Q.10 Strain gauge converts _____ into electrical quantity.

Q.11 A potentiometer is _____ terminal resistor.

Q.12 Inductive microphone works on the principle of magnetic induction to generate audio signal. (True/False)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Define the following

- a) Time constant
- b) Hysteresis
- c) Dead time
- d) Response time

Q.14 Explain the working of potentiometer with diagram.

Q.15 What are the advantages and disadvantages of Strain gauge?

Q.16 Explain the working of potentiometer with diagram.

Q.17 Write down any four advantages of LVDT.

Q.18 Write down any four applications of capacitive pick-up.

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